

- **Watch for:** Radiator fit (front/top), pump noise over time, need for good intake/exhaust balance.
- **Who it suits:** High-end CPUs, warm rooms, compact cases needing low-profile CPU area.

Custom water loops

- **Pros:** Best thermals/noise potential, GPU + CPU cooling, showcase aesthetics.
- **Watch for:** Cost, planning, maintenance, risk if rushed.
- **Who it suits:** Enthusiasts who want bespoke performance and showpiece builds.

Indian climate notes (hot + dusty reality)

- **Upsize cooling** in rooms hitting 32–40 °C. Favor twin-tower air or 240/360 mm AIO; skip tiny downdraft coolers.
- **Prefer mesh-front cases** with dust filters and at least 2× front intakes.
- **Clean filters** every 1–2 months; a 3–5 °C temp drop after cleaning is common.
- **Room matters:** A small, closed room heat-soaks. Improve room airflow or run a fan/AC during heavy work.

Case fundamentals: form factor, airflow, features

A good case simplifies the build, improves temps, and looks the part.

Form factors (what fits where)

- **ATX mid-tower:** The default sweet spot. Fits tall coolers,

long GPUs, 240/280/360 mm radiators.

- **Micro-ATX tower:** Slightly smaller; still practical. Fewer expansion slots, watch radiator support.
- **Mini-ITX (SFF):** Small footprint, portability. Demands careful part choices (SFX PSU, short GPU, low-profile or AIO).
- **E-ATX full tower:** Space galore for custom loops, multi-rad, storage-heavy builds.

Airflow vs aesthetics (mesh, glass, silence)

- **Mesh front/top:** Best airflow and temps. Slightly louder, more dust—filters required.
- **Glass/closed fronts:** Sleeker and quieter but warmer. Look for side vents or “Flow” variants.
- **Silent cases:** Solid panels + damping foam + fewer vents = quieter at the cost of a few degrees. Pair with bigger, slower-spinning fans.

Tip: The “modern default” is mesh front + glass side: balanced thermals and a clean view inside.

Fans, radiators, and pressure – quick rules that work

- **Direction:** Front/bottom = intake; rear/top = exhaust.
- **Positive pressure:** More intake CFM than exhaust reduces dust.
- **Fan size:** 140 mm moves same air at lower RPM (quieter) vs 120 mm.
- **Top radiators:** Great exhaust path for AIOs; check RAM/VRM clearance.

Cooling	Performance headroom	Noise potential	Maintenance	Space & fit	Cost	Best for
Air cooler	Medium → High (twin-tower)	Low at idle, modest under load	Dusting only	Check cooler height/RAM	₹ → ₹₹	Most gaming/creator PCs
AIO (240/280/360)	High → Very high	Low–medium; pump adds faint hum	Dust filters + occasional fan/rad clean	Check radiator mounts	₹₹ → ₹₹₹	Hot CPUs, warm climates
Custom loop	Very high (CPU & GPU)	Excellent if well-tuned	Periodic coolant service	Needs planning space	₹₹₹+	Flagship, showcase rigs

Case size	Typical motherboard	Cooling support	Build difficulty	Use cases
Mini-ITX (SFF)	Mini-ITX	120/240 mm AIO, few fans	High (tight space)	Minimal desks, LAN rigs
Micro-ATX	mATX/ITX	240/280 mm AIO, 2–4 fans	Medium	Budget, compact yet capable
ATX mid-tower	ATX/mATX/ITX	240/280/360 mm, many fans	Low	Mainstream gaming/creator
E-ATX full tower	E-ATX/ATX	Multi-360 mm, custom loops	Medium (big/heavy)	Flagship, showcase builds